The Symmetric and Antisymmetric Halves

One decomposition, from Babylon to the geometric product

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Roadmap

I	The move you already made	even and odd; half-sum and half-difference
II	The split for matrices	$G = S + A$, and how much room each half has
III	Attention, revisited	which half is a choice and which is not
IV	What the fold destroys	why HRR binding forgets the order
V	Keeping both halves	the geometric product, derived
VI	Close	where the symbol lives

Last session ended on a decomposition we used once and moved past: $G = S + A$. Today it is the subject rather than a step.

Running underneath: **if meanings are vectors and the operations on them are just arithmetic, where does anything symbolic come from?** The two halves turn out to answer different parts of that question.

The Question Underneath Today's Arithmetic

A vector of 300 floats is not a symbol. A dot product is not an inference. Yet we are asking these objects to carry predicate-argument structure. Make the difficulty concrete:

John loves Mary. versus *Mary loves John.*

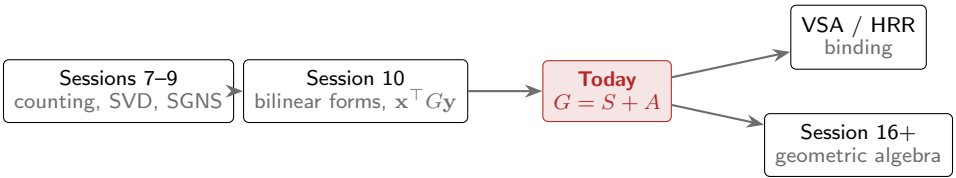
Same words, same lemmas, same bag of context counts. Every representation built so far in this course assigns these two sentences the same vector – not approximately, *identically*, as we proved for the co-occurrence matrix.

Fodor and Pylyshyn's challenge (1988)

A system that can represent loves(j, m) must, by the same means, be able to represent loves(m, j) and tell them apart. They argued no distributed representation does this. Smolensky's answer, and ours today, is that the capacity has a precise location.

Hold onto one word: **order**. The difference between those two sentences is not which concepts are present but *which argument slot each one fills*. By the end of today you will know exactly which half of a bilinear form can see that, and which half is structurally blind to it.

Where We Are



Session 10 built the machinery of $\mathbf{x}^\top G\mathbf{y}$ and spent most of its time on which choices of G behave like a notion of similarity. Near the end it did something different for one slide: it *took* G *apart*.

Today's claim

That one slide was the general method, and the two units still ahead of us both turn on it. Binding loses something, and the geometric product is built to keep it – and these are the same something.

Nothing today requires new machinery. Everything follows from one two-line observation and the discipline of applying it in five places.

Three Places, One Move

You have now seen the same operation three times without it being named.

Where	**The whole**	**Split into**
Session 10, attention	$G = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$	$\left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right) + \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$
Calculus, a function	e^x	$\cosh x + \sinh x$
Grade school, two numbers	7 and 3	$5 + 2$ and $5 - 2$

In every row, one object became **a part that does not care about order** plus **a part that is nothing but order**.

The claim for today: this is one method, it always works, the answer is always unique, and the two halves always answer different questions. Three of the things this course still has to build are instances of it.

Even and Odd, the Case You Know

A function is **even** if $f(-x) = f(x)$ and **odd** if $f(-x) = -f(x)$. Most functions are neither. Every function is a sum of one of each:

$$f(x) \;=\; \underbrace{\frac{f(x)+f(-x)}{2}}_{\text{even part}} + \underbrace{\frac{f(x)-f(-x)}{2}}_{\text{odd part}}$$

Check it: the two terms add to $f(x)$, because the $f(-x)$ cancels. That is the entire proof.

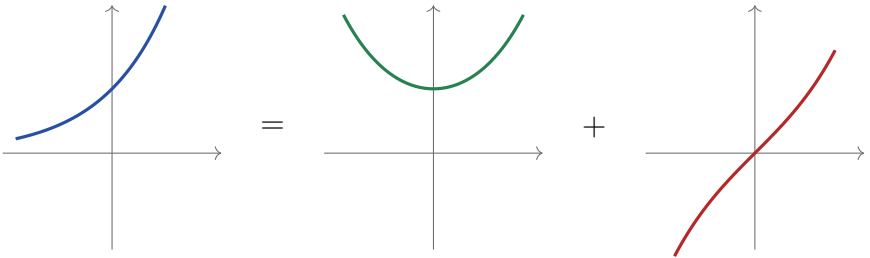
Worked. $f(x) = e^x$, so $f(-x) = e^{-x}$:

$$\frac{e^x + e^{-x}}{2} = \cosh x, \qquad \frac{e^x - e^{-x}}{2} = \sinh x, \qquad e^x = \cosh x + \sinh x$$

The hyperbolic functions are not a separate invention. They are the two halves of the exponential, and that is the whole reason they exist.

Test at $x = 1$: $\cosh 1 = 1.5431$, $\sinh 1 = 1.1752$, sum $= 2.7183 = e$. ✓

The Same Split, Drawn



The even half is what survives reflecting the graph left-to-right; the odd half is what survives rotating it $180°$. Neither picture is a distortion of e^x: together they reconstruct it exactly.

Keep this image. When we split a matrix, S is the part that survives reflecting across the diagonal, and A is the part that survives reflecting and then negating.

The Ingredient: an Involution

What made the split possible was not anything about functions. It was that $x \mapsto -x$ **undoes itself**. An operation σ with $\sigma(\sigma(u)) = u$ is an **involution**, and given one, every u splits uniquely:

$$u \;=\; \underbrace{\tfrac{1}{2}\big(u + \sigma(u)\big)}_{\sigma\text{-even}} \;+\; \underbrace{\tfrac{1}{2}\big(u - \sigma(u)\big)}_{\sigma\text{-odd}}$$

Setting	**The involution σ**	**Check it undoes itself**
functions on $\mathbb{R}$	$f(x) \mapsto f(-x)$	$-(-x) = x$
pairs of numbers	$(x, y) \mapsto (y, x)$	swapping twice
square matrices	$G \mapsto G^\top$	$(G^\top)^\top = G$
complex numbers	$z \mapsto \bar{z}$	$\bar{\bar{z}} = z$
products of two things	$\mathbf{ab} \mapsto \mathbf{ba}$	swapping twice

Five rows, one theorem. The rest of this session is the last three rows worked out, and we will not need to re-prove anything.

Why Halves, and Why the Answer Is Forced

Why the 2. You want $g(x) + h(x) = f(x)$ with g even and h odd. Evaluate at $-x$ as well:

$$g(x) + h(x) = f(x), \qquad g(x) - h(x) = f(-x)$$

Two equations, two unknowns. Add them and h drops out; subtract and g drops out. The halves come from solving a 2×2 system, not from a convention.

Uniqueness

Those two equations have exactly one solution, so there is exactly one way to write f as even plus odd. The decomposition is not a choice you make. It is a fact about f.

This matters later: when we split a matrix, or a product, we are not proposing a convenient coordinate system. We are reading off something the object already had.

The only ingredient was an operation that undoes itself when applied twice: $x \mapsto -x$ here, transposition shortly, argument-swapping after that. **Anywhere you have such an involution, you have this split.**

The Babylonian Version

The oldest form of this move is about four thousand years old, and it is the same algebra.

Old Babylonian scribes solved *find two numbers given their sum and product* by taking the **half-sum** and the **half-difference**. Any pair (x, y) is

$$x = m + d, \qquad y = m - d, \qquad \text{where} \qquad m = \frac{x+y}{2}, \quad d = \frac{x-y}{2}$$

Worked. $x + y = 10$ and $xy = 21$. Then $m = 5$, and

$$xy = (m+d)(m-d) = m^2 - d^2 = 21 \quad \Longrightarrow \quad d^2 = 25 - 21 = 4 \quad \Longrightarrow \quad d = 2$$

So $x = 7$, $y = 3$. $7 + 3 = 10$, $7 \times 3 = 21$. ✓

m is the symmetric part: swap x and y and it does not move. d is the antisymmetric part: swap them and it changes sign. The trick works because the product sees only m^2 and d^2, so the sign of d never matters.

The same device is the completion of the square, the $\pm$ in the quadratic formula, and the midpoint-and-radius form of an interval.

The Same Two Lines for a Matrix

The involution is now transposition: applied twice it returns you to where you started.

Definition. G is **symmetric** if $G^\top = G$, **antisymmetric** if $G^\top = -G$. Every square G splits uniquely:

$$G \;=\; \underbrace{\tfrac{1}{2}\big(G + G^\top\big)}_{S} \;+\; \underbrace{\tfrac{1}{2}\big(G - G^\top\big)}_{A}$$

Same proof as before: $S + A = G$ by cancellation, $S^\top = S$ and $A^\top = -A$ by inspection, uniqueness from the same 2×2 system with $G^\top$ in place of $f(-x)$.

Worked, on a lexical relation. Let $E_{ij} = 1$ when word i *entails* word j, over $\{\text{dog}, \text{animal}\}$. Both entail themselves, dog entails animal, and animal does not entail dog:

$$E = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \quad\Longrightarrow\quad S = \begin{pmatrix} 1 & \tfrac{1}{2} \\ \tfrac{1}{2} & 1 \end{pmatrix}, \qquad A = \begin{pmatrix} 0 & \tfrac{1}{2} \\ -\tfrac{1}{2} & 0 \end{pmatrix}$$

Read the halves as English. S says dog and animal are *related*, strength $\frac{1}{2}$, in neither direction particularly – what every similarity measure in this course reports. A says the arrow runs dog $\to$ animal and not back.

The Split, Drawn on the Entries

Transposition reflects the grid across its main diagonal. S averages each cell with its mirror image; A takes half their difference.



The diagonal of G passes into S untouched and contributes nothing to A, because a cell is its own mirror image there. That single observation is the proof of two separate facts on the next slide.

What Each Half Can See

Two facts that do all the work later. Both are one line.

The antisymmetric half is invisible to lengths: $\mathbf{x}^\top A\mathbf{x} = 0$ for every $\mathbf{x}$.

Why. It is a number, so it equals its own transpose: $\mathbf{x}^\top A\mathbf{x} = \mathbf{x}^\top A^\top \mathbf{x} = -\mathbf{x}^\top A\mathbf{x}$. A number equal to its own negative is zero.

The symmetric half is invisible to order: $\mathbf{x}^\top S\mathbf{y} = \mathbf{y}^\top S\mathbf{x}$, while $\mathbf{x}^\top A\mathbf{y} = -\mathbf{y}^\top A\mathbf{x}$.

	one word	**two words**	**lexical reading**
symmetric S	everything it has	how alike	relatedness, similarity
antisymmetric A	nothing at all	which way	hypernymy, entailment, precedence

In the entailment matrix: every word entails *itself*, and that reflexive fact sits entirely in $S - A$ had zeros on its diagonal. So no measurement of a single word, however rich, can tell you anything about the direction of a relation it participates in. Direction is irreducibly a two-word fact.

How Much Room Each Half Has

Count the free entries. A symmetric $n \times n$ matrix is determined by its diagonal and one triangle; an antisymmetric one has a forced zero diagonal and one triangle.

$$\dim \mathrm{Sym}(n) = \frac{n(n+1)}{2}, \qquad \dim \mathrm{Skew}(n) = \frac{n(n-1)}{2}, \qquad \text{sum} = n^2$$

n	2	3	4	8
symmetric	3	6	10	36
antisymmetric	1	3	6	28
all matrices	4	9	16	64

Over a vocabulary, $\binom{n}{2}$ is **the number of word pairs**, and the antisymmetric part has one degree of freedom per pair – because for each pair there is exactly one thing a directed relation can say: *which way, and how strongly*. Three words, three arrows; ten words, forty-five. It is also one per pair of axes, that is, one per plane – remember that.

The Plane Is a Special Case

Look again at $n = 2$. The antisymmetric matrices form a *one-dimensional* space: every one of them is a multiple of

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

Consequence. The "which way" content of *any* G on $\mathbb{R}^2$ is $\lambda \cdot \mathbf{x}^\top J \mathbf{y}$ for a single number λ. There is nothing to choose but a scale – and in lexical terms that is obvious: with two words there is one pair, so there is one arrow, and the only questions are which way it points and how strongly. Our entailment matrix had $\lambda = \frac{1}{2}$.

The symmetric half is nothing like that. It is a choice from a *three*-dimensional space, and last session toured exactly those choices: I, $\mathrm{diag}(4, 1)$, the correlated $\left(\begin{smallmatrix} 1 & .8 \\ .8 & 1 \end{smallmatrix}\right)$, the terrain metric, and the one that was not a metric at all.

So a bilinear form on the plane is **a metric you pick** plus **an orientation you can only scale**. A two-dimensional accident – at $n = 3$ the antisymmetric space is already three-dimensional – but it is why the next few slides can compare an attention head with a wedge product and find literally the same number.

Which Half Has This Course Been Using?

Every similarity measure in Sessions 7 through 10 lives in the symmetric half. Worth being explicit, because it means none of them could ever have done what Part III needs.

From earlier in the course	Needs	Why
cosine, Jaccard, Dice	only S	symmetric by definition
PMI and PPMI, Session 7	only S	$\mathrm{PMI}(w, c) = \mathrm{PMI}(c, w)$
$\|\mathbf{x}\|_G$, the unit ball, $\cos_G$	only S	A contributes 0
positive definiteness, Cholesky	only S	statements about $\mathbf{x}^\top G\mathbf{x}$
hypernymy, entailment	only A	the arrow
attention's directionality	only A	the half that flips sign

Look at the PMI row. You proved $\mathrm{PMI}(w, c) = \mathrm{PMI}(c, w)$ in a Session 7 exercise, and were asked what that symmetry rules out as a model of entailment. The answer, now sayable precisely: a symmetric quantity has no A at all, so it has nothing with which to record a direction – not weakly, not approximately, not at all.

So "Is G a Metric?" Was Never About All of G

Last session asked which choices of G behave like a notion of similarity. Given the table on the previous slide, that question was always about S alone, with the tacit assumption $A = 0$.

Two consequences worth stating

Every ellipse you drew was the ellipse of the symmetric half. A non-symmetric G and its symmetric part S draw the *same* unit ball, so the picture could never have shown you the difference.

To see A you must compare two different things. Lengths, norms, and self-similarity are all blind to it, by the result two slides back.

Put semantically: a vector space with a metric is a theory of *how alike words are*. It is not, and cannot be extended into, a theory of how they are *ordered* – by generality, by entailment, by time, or by argument position.

That is not a defect in metrics. It is a statement about which of the two halves a metric is made of, and the rest of today is about the other one.

Half the Answer: Symbols Must Be Distinguishable

Before structure, a symbol system needs **identity**: dog must be tellable from cat and from every other symbol. In a vector space that means *orthogonality*, and orthogonality is a question you ask a bilinear form:

$$\mathbf{x} \perp_G \mathbf{y} \quad \text{iff} \quad \mathbf{x}^\top G \mathbf{y} = 0$$

That relation had better be symmetric – "dog is distinct from cat" cannot hold in one direction only. Since $\mathbf{x}^\top A \mathbf{y} = -\mathbf{y}^\top A \mathbf{x}$, a well-defined notion of distinctness lives entirely in S.

And it is nearly free, if you have dimensions to spare

Two random unit vectors in $\mathbb{R}^d$ have expected cosine 0 and standard deviation $1/\sqrt{d}$. At $d = 10{,}000$ that is 0.01: a few thousand randomly chosen vectors are mutually near-orthogonal with overwhelming probability, with no design effort at all.

This is why vector symbolic architectures are *hyper*dimensional: those dimensions buy a large alphabet of reliably distinguishable symbols out of randomness. So the symmetric half supplies identity. It cannot supply structure, and the rest of the session is about what can.

The Co-occurrence Matrix Has No Odd Half

Apply the test to the first matrix of the course. A symmetric-window co-occurrence count over $\{\text{dog}, \text{bites}, \text{man}\}$:

$$\textit{dog bites man} \longrightarrow \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \qquad \textit{man bites dog} \longrightarrow \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$$

Identical. The matrix is symmetric by construction, so $A = 0$ exactly, and the two sentences are indistinguishable to everything built on top of it.

That is not a flaw in the counting code. A window of $\pm k$ is a *symmetric* notion of context, so the object it produces lives entirely in $\mathrm{Sym}(n)$, and no weighting, factorization, or dimension reduction downstream can recover what was never recorded.

PPMI is entrywise, so it preserves symmetry. The SVD of a symmetric matrix is symmetric. **Every representation in Sessions 7 and 8 is order-blind, and the decomposition says why in one word.**

This is the structural statement behind the order test in Problem Set 2, where boathouse and houseboat received identical predictions under addition and multiplication.

In-Class Exercise 1

1. Split $M = \left(\begin{smallmatrix} 4 & 6 \\ 2 & 0 \end{smallmatrix}\right)$ into $S + A$.
2. Using your S and A, compute $\mathbf{x}^\top M \mathbf{y}$ and $\mathbf{y}^\top M \mathbf{x}$ for $\mathbf{x} = (1, 2)$, $\mathbf{y} = (3, 1)$ *without* multiplying by M directly. What is the relationship between the two answers?
3. A matrix is both symmetric and antisymmetric. Which matrix is it?
4. $\star$ The diagonal of an antisymmetric matrix is forced to be zero. Show it from $A^\top = -A$ in one line, and say what that fact is in the language of the previous slide.

Five minutes. Item 4 is the starred one.

In-Class Exercise 1: Answers

1. $M^\top = \left(\begin{smallmatrix} 4 & 2 \\ 6 & 0 \end{smallmatrix}\right)$, so $S = \left(\begin{smallmatrix} 4 & 4 \\ 4 & 0 \end{smallmatrix}\right)$ and $A = \left(\begin{smallmatrix} 0 & 2 \\ -2 & 0 \end{smallmatrix}\right)$.

2. $\mathbf{x}^\top S\mathbf{y} = 4(3) + 4(1) + 4(6) + 0 = 40$ and $\mathbf{x}^\top A\mathbf{y} = 2(1)(1) - 2(2)(3) = -10$.

$$\mathbf{x}^\top M\mathbf{y} = 40 - 10 = 30, \qquad \mathbf{y}^\top M\mathbf{x} = 40 + 10 = 50$$

They share a symmetric part and differ by twice the antisymmetric one. Check directly: $M\mathbf{y} = (18, 6)$, and $(1, 2) \cdot (18, 6) = 30$. ✓

3. Only the zero matrix. If $G^\top = G$ and $G^\top = -G$ then $G = -G$, so $2G = 0$. The two spaces meet only at 0, which is why the dimensions add to exactly n^2 with nothing double-counted.

4. ⋆ $A_{ii} = -A_{ii}$ forces $A_{ii} = 0$. In the previous slide's language: the diagonal is what a form reports about a single vector paired with itself, and the antisymmetric half reports nothing there. This is $\mathbf{x}^\top A\mathbf{x} = 0$ read one basis vector at a time.

The Attention Score, Split Again

From last session, with $\mathbf{x}_1 = (3, 1)$ and $\mathbf{x}_2 = (1, 3)$ and $G = W_Q^\top W_K = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$:

	$1 \to 2$	$2 \to 1$	
$\mathbf{x}^\top S\mathbf{y}$	16	16	how alike
$\mathbf{x}^\top A\mathbf{y}$	$+8$	-8	which way
total	24	8	

Read the two rows as answers to different questions. The 16 says how much these two tokens have in common and is the same no matter which way you ask. The ± 8 says nothing about commonality and everything about which of them is doing the attending.

Neither row alone is the score. Attention needs both, and the softmax sees only their sum.

Which Half Did the Head Choose?

Put the dimension count beside the split and the two halves stop looking symmetrical in status.

	room to choose at $n = 2$	**what training decides**
symmetric S	3 dimensions	a whole notion of similarity
antisymmetric A	1 dimension	one number: how much, and which sign

The asymmetry of the asymmetry

What a head learns about *similarity* is a genuine choice among many. What it learns about *direction*, in the plane, is a coefficient on the one antisymmetric form that exists.

Do not over-read this. At a realistic $d_k = 64$ the counts are 2080 and 2016 – nearly equal – so the lopsidedness is an artifact of working in two dimensions. The structural point survives at every dimension: the halves are independent, and a model that ties $W_Q = W_K$ sets the second one to zero.

What a Symmetric Head Could Not Do

Suppose a model were built with $W_Q = W_K$, so $G = W_Q^\top W_Q$ is symmetric by construction. A reasonable-looking design, and a crippled one.

	our G	**symmetric** G
score $1 \to 2$	24	16
score $2 \to 1$	8	16
token 1's attention to itself	0.003	0.499
token 1's attention to token 2	**0.996**	0.499

A symmetric head cannot break a tie between two equally similar tokens, because for it "equally similar" is the only fact there is.

Grammatical relations are directional almost everywhere: a verb and its subject, a modifier and its head, a pronoun and its antecedent. Each needs the half a symmetric design throws away. Separate W_Q and W_K is usually explained as more parameters; the sharper statement is that it is what buys the antisymmetric half at all.

What the Odd Half Is Doing in a Sentence

Take the two sentences from the start of the session and parse them:



The *edges* are identical as unordered pairs: $\{\text{loves}, \text{John}\}$ and $\{\text{loves}, \text{Mary}\}$ both times. Everything that distinguishes the two readings is in the arrowheads – which end of each edge is the head and which the dependent.

An unordered pair is what a symmetric form reports. An ordered pair is what an antisymmetric one reports. A dependency parse is, in this precise sense, a record of the odd half.

So when a head scores loves $\to$ John at 24 and John $\to$ loves at 8, the asymmetry is not an artifact. It is the only place in the arithmetic where an arrowhead could possibly be stored.

Many Heads, Many Splits

A layer with H heads carries H separate matrices $G_h = W_{Q,h}^\top W_{K,h}$, and each one splits on its own.

$$G_h \;=\; S_h + A_h, \qquad h = 1, \ldots, H$$

A head with	**behaves like**	
$A_h \approx 0$	a similarity measure	symmetric, order-blind
$S_h \approx 0$	a pure orientation detector	scores $\pm$ by direction alone
both substantial	similarity gated by direction	the general case

Nothing in the architecture forces a head toward any of these. The ratio $\|A_h\|_F / \|S_h\|_F$ is a single number per head, computable from the weights alone, and it says how much of what the head does is directional.

What heads in trained models actually look like on this measure is an empirical question, and the literature on what individual heads learn is contested enough that I would not assert a pattern here. It is, however, a clean thing to go and measure, and the computation is three lines.

Binding, and the Problem With It

Recall the binding operation from the VSA session. To bind a role vector $\mathbf{r}$ to a filler vector $\mathbf{f}$, the honest answer is the outer product:

$$\mathbf{r}\mathbf{f}^\top \in \mathbb{R}^{n\times n} \qquad \text{which is } \textbf{not} \quad \mathbf{f}\mathbf{r}^\top$$

Order is kept, and that is the point of binding. The cost is n^2 numbers for every pair, which does not scale.

Holographic Reduced Representations fix the size by **folding**: sum the outer product along its wrapped anti-diagonals, giving n numbers again. That fold is circular convolution.

$$(\mathbf{r} \circledast \mathbf{f})_k \;=\; \sum_i r_i \, f_{(k-i) \bmod n}$$

The thing to notice

Circular convolution is **commutative**: $\mathbf{r} \circledast \mathbf{f} = \mathbf{f} \circledast \mathbf{r}$. The operation we adopted to keep order does not keep order.

The next two slides say exactly which part went missing, and it is not a coincidence.

The Fold, By Hand

$\mathbf{u} = (1, 2, 0, 1)$ and $\mathbf{v} = (2, 0, 1, 1)$ in $\mathbb{R}^4$. The outer product, with each cell labeled by $k = (i + j) \bmod 4$:

$\mathbf{u}$ \ $\mathbf{v}^\top$	2	0	1	1
1	4^{1}	0^{2}	2^{3}	2^{0}
2	0^{2}	0^{3}	0^{0}	0^{1}
0	2^{3}	0^{0}	1^{1}	1^{2}
1	2^{3}	0^{0}	1^{1}	1^{2}

shaded: the four products with $k = 1$,
$0 + 4 + 0 + 1 = 5$

$$\mathbf{u} \circledast \mathbf{v} = (4,\ 5,\ 2,\ 5) \qquad \mathbf{v} \circledast \mathbf{u} = (4,\ 5,\ 2,\ 5)$$

Identical, although $\mathbf{u}\mathbf{v}^\top \neq \mathbf{v}\mathbf{u}^\top$. Sixteen numbers went in, four came out, and whatever distinguished the two orders was in the twelve that did not survive.

Which Twelve: the Fold Annihilates A

Split the outer product before folding, $\mathbf{u}\mathbf{v}^\top = S + A$, and fold each half on its own:

what is folded	result	
$\mathbf{u}\mathbf{v}^\top$	$(4, 5, 2, 5)$	the binding
symmetric half S	$(4, 5, 2, 5)$	all of it
antisymmetric half A	$(0, 0, 0, 0)$	none of it

Why, in one line. Folding adds the cells on each wrapped anti-diagonal, and transposing permutes every anti-diagonal onto itself. So A folds to the same thing as $A^\top = -A$, and a quantity equal to its own negative is zero.

Circular convolution is exactly **the symmetric half of the outer product**, folded. Commutativity is not a quirk of the formula: the operation is blind to order because the order-carrying half is precisely what the fold removes.

What This Costs, and Where We Take It Up

Fold n^2 numbers into n and two different things are lost. Keeping them apart matters, because only one of them is fixable.

Loss	Character	Does more dimension help?
antisymmetric half annihilated	exact, total, structural	no – it was multiplied by zero
symmetric half compressed to n	approximate, noisy	yes – crosstalk falls as $1/\sqrt{n}$

At $n = 1000$ the fold first discards $499{,}500$ numbers that encode nothing but argument order, then compresses the remaining $500{,}500$ into 1000. The VSA literature's capacity analysis – how many pairs can you superpose before retrieval degrades – is entirely about the second row.

So "unbinding is approximate" was doing double duty: one sense improves with dimension, the other says argument order is *not present*. That is why HRR stipulates $\mathbf{r}_{\text{agent}}$ and $\mathbf{r}_{\text{theme}}$ as distinct vectors in advance – the binding operation cannot tell you which slot you are in.

The repairs are the subject of the next session, which VSA has earned on its own. Today supplies the diagnosis they all answer.

Stop Throwing One Away

Every product we have used on two vectors discards a half.

Product	**Keeps**	**Discards**
dot product $\mathbf{a} \cdot \mathbf{b}$	symmetric	all orientation
circular convolution $\mathbf{a} \circledast \mathbf{b}$	symmetric	all orientation
wedge $\mathbf{a} \wedge \mathbf{b}$	antisymmetric	all magnitude agreement
outer product $\mathbf{a}\mathbf{b}^\top$	both	nothing, but costs n^2

The outer product already keeps everything. What it does not do is stay inside a workable algebra: you cannot multiply two of its outputs and get the same kind of thing back.

The move. Define a product $\mathbf{ab}$ and *demand* that its symmetric half be the dot product and its antisymmetric half be the wedge. Then see what you are committed to.

The reverse of the usual presentation: not defining a product and noticing it splits, but specifying the halves and letting them determine the product.

The Geometric Product, Derived

Write $\mathbf{ab}$ for the product we want. By the decomposition, whatever it is, it satisfies

$$\mathbf{ab} \;=\; \underbrace{\tfrac{1}{2}\big(\mathbf{ab}+\mathbf{ba}\big)}_{\text{symmetric half}} \;+\; \underbrace{\tfrac{1}{2}\big(\mathbf{ab}-\mathbf{ba}\big)}_{\text{antisymmetric half}}$$

Now name the two halves as we demanded:

$$\mathbf{a}\cdot\mathbf{b} \;\coloneqq\; \tfrac{1}{2}\big(\mathbf{ab}+\mathbf{ba}\big), \qquad \mathbf{a}\wedge\mathbf{b} \;\coloneqq\; \tfrac{1}{2}\big(\mathbf{ab}-\mathbf{ba}\big)$$

The geometric product

$$\mathbf{ab} \;=\; \mathbf{a}\cdot\mathbf{b} \;+\; \mathbf{a}\wedge\mathbf{b}$$

That equation is usually given as a definition to accept. It is not: it is the even/odd decomposition of a product, with the two halves given their names. The involution here is swapping the arguments, exactly as $x \mapsto -x$ was in Part I.

Nothing has been assumed about what $\mathbf{ab}$ *is* yet. Only that it exists and that its two halves are the two products we already have.

Worked, With Numbers You Have Seen

Take the same two vectors as the attention example: $\mathbf{a} = 3\mathbf{e}_1 + \mathbf{e}_2$, $\mathbf{b} = \mathbf{e}_1 + 3\mathbf{e}_2$ in the plane.

symmetric half	$\mathbf{a} \cdot \mathbf{b} = 3(1) + 1(3) = 6$	a scalar
antisymmetric half	$\mathbf{a} \wedge \mathbf{b} = \big(3(3) - 1(1)\big)\,\mathbf{e}_1\mathbf{e}_2 = 8\,\mathbf{e}_1\mathbf{e}_2$	a bivector
geometric product	$\mathbf{ab} = 6 + 8\,\mathbf{e}_1\mathbf{e}_2$	
reversed	$\mathbf{ba} = 6 - 8\,\mathbf{e}_1\mathbf{e}_2$	

That 8 is not a coincidence. It is the same 8 the attention head produced as $\mathbf{x}^\top A\mathbf{y}$, three slides ago. It has to be: in the plane the antisymmetric forms are one-dimensional, so *every* antisymmetric bilinear form on $\mathbb{R}^2$ is a multiple of this one.

The head's symmetric half gave 16 and the Euclidean dot product gives 6, because those are different choices from a three-dimensional space of metrics. The directional halves agree exactly, because there was nothing to choose.

8 is also the signed area of the parallelogram on $\mathbf{a}$ and $\mathbf{b}$, and the determinant $\left|\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right|$. All the same number.

What the Basis Vectors Have to Do

Apply the two halves to basis vectors and the multiplication rules fall out with no further input.

A vector with itself. $\mathbf{e}_1 \wedge \mathbf{e}_1 = 0$ since the odd half vanishes on equal arguments, and $\mathbf{e}_1 \cdot \mathbf{e}_1 = 1$. So

$$\mathbf{e}_1\mathbf{e}_1 \;=\; 1, \qquad \text{and likewise} \qquad \mathbf{e}_i\mathbf{e}_i = 1$$

Two different basis vectors. $\mathbf{e}_1 \cdot \mathbf{e}_2 = 0$ because they are orthogonal, so the product is pure odd:

$$\mathbf{e}_1\mathbf{e}_2 \;=\; \mathbf{e}_1 \wedge \mathbf{e}_2 \qquad \Longrightarrow \qquad \mathbf{e}_1\mathbf{e}_2 = -\,\mathbf{e}_2\mathbf{e}_1$$

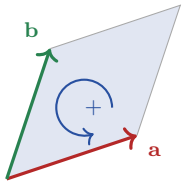
because the odd half changes sign when you swap. Write $\mathbf{e}_{12}$ for this object.

And one surprise, for free

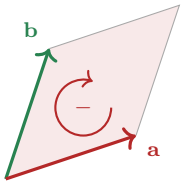
$\mathbf{e}_{12}^2 = \mathbf{e}_1\mathbf{e}_2\mathbf{e}_1\mathbf{e}_2 = -\mathbf{e}_1\mathbf{e}_1\mathbf{e}_2\mathbf{e}_2 = -1$, using anticommutation once and $\mathbf{e}_i\mathbf{e}_i = 1$ twice.

An object squaring to -1 has appeared without anyone postulating it. That is where the complex numbers come from next session, and the two lines above are the whole of $\mathrm{Cl}(2,0)$'s multiplication table.

What a Bivector Is



$\mathbf{a} \wedge \mathbf{b} = +8\,\mathbf{e}_{12}$



$\mathbf{b} \wedge \mathbf{a} = -8\,\mathbf{e}_{12}$

Same two vectors, same parallelogram, same area 8. What differs is the **sense** in which you traverse it: $\mathbf{a}$ then $\mathbf{b}$, or $\mathbf{b}$ then $\mathbf{a}$.

A bivector is an *oriented area*: a plane, a magnitude, and a circulation. It is not a vector and does not point anywhere. Compare a vector, which is an oriented length.

This is the geometric content of the sign flip that the attention head was using to tell $i \to j$ from $j \to i$.

Scalar Plus Bivector Is Not Strange

The product of two vectors came out as a scalar plus an object of another kind. That looks wrong the first time. It is the same shape as everything in this session:

	symmetric part	**antisymmetric part**
e^x	$\cosh x$	$\sinh x$
a pair of numbers	half-sum m	half-difference d
a matrix G	S	A
an attention score	16	± 8
$\mathbf{ab}$	scalar $\mathbf{a} \cdot \mathbf{b}$	bivector $\mathbf{a} \wedge \mathbf{b}$

In every row the two halves are different kinds of thing, kept side by side rather than added into one number. $\cosh x + \sinh x$ is no more homogeneous than $6 + 8\mathbf{e}_1\mathbf{e}_2$; we are simply used to the first.

Complex numbers are the same arrangement again: a real part blind to conjugation, an imaginary part that flips sign under it. Next session shows $\mathrm{Cl}(\mathbb{R}^2)$ *contains* the complex numbers, and this is why.

What the Bivector Count Was Telling You

The antisymmetric matrices in n dimensions had dimension $\binom{n}{2}$ – one for each pair of axes.

n	2	3	4	8
antisymmetric forms $\binom{n}{2}$	1	3	6	28
planes spanned by pairs of axes	1	3	6	28
basis bivectors in $\mathrm{Cl}(n,0)$	1	3	6	28

These are the same count because they are the same objects. An antisymmetric bilinear form on $\mathbb{R}^n$ *is* a bivector, written in matrix coordinates.

In three dimensions this is why the cross product seems to work and then stops working. $\binom{3}{2} = 3$ happens to equal 3, so bivectors in $\mathbb{R}^3$ can be disguised as vectors. In $\mathbb{R}^4$, $\binom{4}{2} = 6 \neq 4$, and the disguise fails. The wedge keeps working; the cross product does not generalize.

Session 16 builds the full multiplication table, including what happens when bivectors multiply each other.

In-Class Exercise 2

1. Compute $\mathbf{a} \cdot \mathbf{b}$ and $\mathbf{a} \wedge \mathbf{b}$ for $\mathbf{a} = (2, 1)$ and $\mathbf{b} = (4, 2)$. What is geometrically special about this pair, and which half detected it?
2. From $\mathbf{ab} = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \wedge \mathbf{b}$ and $\mathbf{ba} = \mathbf{a} \cdot \mathbf{b} - \mathbf{a} \wedge \mathbf{b}$, show that $\mathbf{aa}$ is a scalar, and say which scalar.
3. Two vectors are orthogonal. What does their geometric product reduce to? Two vectors are parallel. Same question.
4. $\star$ The fold in Part IV destroyed the antisymmetric half. The dot product also destroys it. Yet we called one of these a defect and the other not. What is the difference?

Six minutes. Item 4 is the starred one, and the one worth arguing about.

In-Class Exercise 2: Answers

1. $\mathbf{a} \cdot \mathbf{b} = 8 + 2 = 10$, and $\mathbf{a} \wedge \mathbf{b} = (2)(2) - (1)(4) = 0$. The vectors are parallel ($\mathbf{b} = 2\mathbf{a}$), they span no area, and it is the antisymmetric half that reports it. The dot product cannot: 10 looks like any other number.

2. Setting $\mathbf{b} = \mathbf{a}$, the antisymmetric half is $\frac{1}{2}(\mathbf{aa} - \mathbf{aa}) = 0$. So $\mathbf{aa} = \mathbf{a} \cdot \mathbf{a} = \|\mathbf{a}\|^2$. A vector squared is its squared length – the single most useful fact in the algebra, and it follows from $\mathbf{x}^\top A\mathbf{x} = 0$.

3. Orthogonal: $\mathbf{a} \cdot \mathbf{b} = 0$, so $\mathbf{ab} = \mathbf{a} \wedge \mathbf{b}$, a pure bivector, and $\mathbf{ab} = -\mathbf{ba}$. Parallel: $\mathbf{a} \wedge \mathbf{b} = 0$, so $\mathbf{ab} = \mathbf{a} \cdot \mathbf{b}$, a pure scalar, and $\mathbf{ab} = \mathbf{ba}$. The product is purely one half or purely the other exactly at the two extremes.

4. $\star$ The dot product announces what it is discarding; its output is a scalar and nobody expects orientation from it. The fold was adopted *in order to* preserve role–filler order and does not do so, while returning an object the same shape as its inputs, which makes the loss easy to miss. A lossy operation is not a defect. An operation that is lossy in exactly the respect it was chosen for is.

Grades: a Place for Each Half

The product of two vectors produced two kinds of object. Name them by how many vectors it takes to build one.

Grade	**Object**	**In** $\mathrm{Cl}(2,0)$	**Count**
0	scalar	1	1
1	vector	$\mathbf{e}_1,\ \mathbf{e}_2$	2
2	bivector	$\mathbf{e}_{12}$	1
	total		$4 = 2^2$

The geometric product of two vectors lands in grades 0 and 2: the symmetric half in grade 0, the antisymmetric half in grade 2. Grade parity is the even/odd distinction of Part I, made into a grading on the algebra.

In n dimensions the grades run 0 to n with $\binom{n}{k}$ objects in grade k, totalling 2^n. Grade 2 has $\binom{n}{2}$ entries – the number counted twice already, as antisymmetric matrices and as planes. A sum across grades is a *multivector*.

Why This Product and Not Another

We asked for a product whose two halves are the dot and the wedge. That request has consequences worth stating before next session relies on them.

Property	Dot	Geometric
output stays in the algebra	no, drops to scalars	yes
associative	not defined	yes
vectors have inverses	no	$\mathbf{a}^{-1} = \mathbf{a}/\|\mathbf{a}\|^2$
keeps orientation	no	yes

The inverse is the one to dwell on. From Exercise 2, $\mathbf{aa} = \|\mathbf{a}\|^2$, so dividing by that scalar gives $\mathbf{a}\left(\mathbf{a}/\|\mathbf{a}\|^2\right) = 1$. You can divide by a vector, which is not available in any product this course has used so far.

Division is what makes the algebra do work rather than merely record structure: it is how reflections and rotations get written as $\mathbf{v} \mapsto \mathbf{ava}^{-1}$. Keeping both halves was not tidiness – it is what buys invertibility.

One Method, Five Places

Object	Involution	Even part	Odd part
function f	$x \mapsto -x$	$\cosh$-like	$\sinh$-like
pair (x, y)	swap	half-sum	half-difference
matrix G	transpose	S, the metric	A, the orientation
outer product $\mathbf{r}\mathbf{f}^\top$	transpose	survives the fold	destroyed by it
product $\mathbf{ab}$	swap arguments	$\mathbf{a} \cdot \mathbf{b}$, scalar	$\mathbf{a} \wedge \mathbf{b}$, bivector

One requirement generates all five: an operation that undoes itself when applied twice. Given that, the split exists, it is unique, and the even half is order-blind while the odd half is nothing but order.

Two more you will meet. In quantum mechanics the anticommutator $\{A, B\} = AB + BA$ and commutator $[A, B] = AB - BA$ are these halves without the $\frac{1}{2}$. In Lie theory the odd half of an associative product is the Lie bracket, and the even half is the Jordan product.

Where the Symbol Lives

Back to the question from the start. A vector is not a symbol and a dot product is not an inference. What a symbol system needs is three capacities, and each has an address.

What symbolic reasoning needs	Where it lives	What supplies it
identity: dog from cat	symmetric half	orthogonality, bought with dimension
structure: loves(j, m) from loves(m, j)	antisymmetric half	the only part that flips sign
inference: compose, then *de*compose	both at once	a product with inverses

No single vector carries any of this. The structure is in the *operations*. Which is why the failures we catalogued all look alike: a bag of words, a co-occurrence matrix, a convolution binding – each keeps identity and discards order, so each can represent the concepts in *John loves Mary* and none can represent the sentence. Fodor and Pylyshyn were pointing at a missing half, not a missing idea.

Three Questions to Ask Any Bilinear Object

A method you can run on anything this course hands you from here on.

1. **What is the involution?** Negation, transposition, conjugation, swapping the arguments. If there is one, there is a split.
2. **Which half does this operation keep?** A norm keeps only the even half. A determinant-like quantity keeps only the odd. A fold may silently keep one and destroy the other.
3. **Does the task need the half that was dropped?** Similarity does not. Order, orientation, entailment, argument structure and rotation all do.

Object	Keeps	Costs
co-occurrence counts	even only	word order, structurally
circular convolution	even only	role–filler order
attention score	both	nothing
geometric product	both	nothing

Two rows of that table were presented as design decisions about efficiency. They are decisions about which half of the information to keep.

Summary

1. Any object with an involution splits uniquely into an even and an odd half. The halves come from solving a 2×2 system, and the decomposition is a fact about the object, not a convention.
2. For matrices, S carries all of the length information and none of the order; A carries none of the length and all of the order. $\mathbf{x}^\top A\mathbf{x} = 0$ always.
3. Separate W_Q and W_K exist to buy the antisymmetric half. Without it a head cannot distinguish i attending to j from the reverse.
4. Circular convolution is the symmetric half of the outer product. HRR binding is commutative because the fold annihilates A exactly, which is why roles must be fixed in advance.
5. The geometric product $\mathbf{ab} = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \wedge \mathbf{b}$ is not a definition to accept. It is this decomposition with the halves named, and it is the one product that keeps both.

Next session opens with the multiplication table of $\mathrm{Cl}(n, 0)$. You already know why it has to have two kinds of entry.

Where This Is Going

Next	Picks up	Which half
Session 16, the geometric product	$\mathbf{ab} = \mathbf{a}\cdot\mathbf{b} + \mathbf{a}\wedge\mathbf{b}$	both, by construction
Bivectors and rotations	$\mathbf{e}_{12}^2 = -1,\ \ \mathbf{ava}^{-1}$	the odd half acting
Signature $\mathrm{Cl}(p,q,r)$	which $\mathbf{e}_i\mathbf{e}_i$ are $+1, -1, 0$	the even half, chosen
Binding without the fold	keep the grade-2 part	the half HRR lost

The last row is the thread worth watching. The complaint we raised against circular convolution – that it is commutative, so it cannot encode which argument slot a filler occupies – is answered by a product that keeps the antisymmetric part instead of folding it away.

Session 10 asked what the middle of $\mathbf{x}^\top G\mathbf{y}$ could be, and answered: anything you declare. Session 16 declares it as a **signature**, which is the symmetric half fixed in advance by saying which basis vectors square to what.

Read for next time: the opening of the geometric algebra notes, through the multiplication table for $\mathrm{Cl}(2,0)$. You have already derived every entry in it.

Exercises

1. Split $\begin{pmatrix} 0 & 3 & 1 \\ -1 & 2 & 0 \\ 5 & 4 & 1 \end{pmatrix}$ into $S + A$ and verify that $\mathbf{x}^\top A \mathbf{x} = 0$ for $\mathbf{x} = (1, 1, 1)$ and for one $\mathbf{x}$ of your choosing.
2. Prove that $\mathrm{Sym}(n)$ and $\mathrm{Skew}(n)$ intersect only at 0, and use that to explain why $\frac{n(n+1)}{2} + \frac{n(n-1)}{2} = n^2$ is not a coincidence.
3. **Programming.** Take random $\mathbf{u}, \mathbf{v} \in \mathbb{R}^{64}$. Compute $\mathbf{u} \circledast \mathbf{v}$ directly, then split $\mathbf{u}\mathbf{v}^\top$ into $S + A$ and fold each half. Report all three. Then measure how well $\mathbf{v}$ can be recovered from $\mathbf{u} \circledast \mathbf{v}$ and from $\mathrm{fold}(S)$, and explain why the two numbers are equal.
4. Compute $\mathbf{ab}$ and $\mathbf{ba}$ for $\mathbf{a} = (1, 2)$, $\mathbf{b} = (3, -1)$. Confirm that their half-sum and half-difference give back $\mathbf{a} \cdot \mathbf{b}$ and $\mathbf{a} \wedge \mathbf{b}$.
5. $\star$ In $\mathbb{R}^3$ the wedge $\mathbf{a} \wedge \mathbf{b}$ has three components, as does the cross product $\mathbf{a} \times \mathbf{b}$, and they carry the same numbers. Find a property of the cross product that the wedge does not share, and say which of the two you would trust in $\mathbb{R}^4$.